

What are LLMs and how do they work?

LLMs have been at the forefront of generative AI's evolution, enabling machines to understand, process, and generate human-like text. These models are trained on massive datasets containing diverse text from the internet, allowing them to learn the intricacies of human language, including syntax, semantics, and context.

At their core, LLMs are neural networks trained to predict the next word or sequence of words in a text based on the context they've already seen. This ability to predict subsequent words allows LLMs to generate coherent, contextually accurate sentences, paragraphs, and even entire documents.

The training process involves feeding a model vast amounts of text data, which it uses to identify patterns, relationships, and structures in language. Over time, these models learn to understand the underlying rules of grammar, the nuances of various dialects, and even subtleties like tone, sentiment, and style.

LLMs are based on a neural architecture called the Transformer, introduced by Vaswani *et al.* in 2017 in a groundbreaking paper, 'Attention Is All You Need'. The Transformer architecture leverages the attention mechanism, which allows the model to weigh the importance of different words or tokens in a sequence. Unlike earlier models, which processed text sequentially (word by word), Transformers analyse the entire input at once, allowing them to capture long-range dependencies and relationships between words more effectively.

Key players in the LLM landscape

The LLM ecosystem has grown rapidly, with both proprietary and open source models making significant strides in performance and capability. Some of the most notable LLMs include:

GPT-3 (Generative Pre-trained Transformer 3): Developed by OpenAI, GPT-3 is one of the most well-known LLMs and has gained significant attention for its ability to generate coherent and contextually accurate text. With 175 billion parameters (the weights that the model learns during training), GPT-3 is capable of understanding and generating human-like responses in natural language, making it ideal for tasks like writing, summarisation, translation, and even programming. However, GPT-3 is not open source, and access to the model is restricted through OpenAI's API, making it available only to users who can afford the costs.

GPT-J: GPT-J is an open source alternative to GPT-3, developed by EleutherAI, a grassroots collective of researchers and developers. It is trained on 6 billion parameters and can generate high-quality text similar to GPT-3, but it is freely available to anyone who wants to use it. GPT-J is part of a larger movement in the AI community to provide powerful LLMs as open source tools, allowing developers and researchers to experiment and build on the model.

GPT-Neo: Another model from EleutherAI, GPT-Neo comes in various sizes, including a 1.3 billion and 2.7 billion parameter version. GPT-Neo is designed to provide an open source alternative to GPT-3 that is accessible to the broader AI community. It is trained on publicly available datasets, making it a more ethical choice for those looking to avoid proprietary systems.

Llama (Large Language Model Meta AI): Developed by Meta (formerly Facebook), Llama is a family of open source LLMs, ranging from small models with 7 billion parameters to larger ones with 65 billion parameters. Llama aims to provide researchers and developers with a high-quality LLM that is lightweight enough to run on a wide range of hardware,

making it a good option for both research and real-world applications.

The role of transformers and attention mechanisms in LLMs

The Transformer architecture is what sets LLMs apart from traditional neural networks. Unlike earlier sequence models like RNNs (recurrent neural networks) or LSTMs (long short-term memory networks), which process input data in a step-by-step sequence, Transformers process all the input data in parallel, making them much faster and more efficient. The attention mechanism within Transformers allows the model to focus on specific parts of the input sequence, giving it the ability to process long-range dependencies without losing important contextual information. Key components of the Transformer architecture include:

Self-attention: This allows the model to weigh the importance of each word in a sequence relative to every other word, enabling it to capture relationships between distant words (e.g., in long paragraphs or sentences).

Positional encoding: Since Transformers don't process input sequentially, positional encoding is used to give the model information about the order of words in a sequence.

Feed-forward networks: After self-attention, the model passes the output through fully connected feed-forward networks to refine the information and learn higher-level abstractions.

Building open source LLMs: Frameworks and tools

As the popularity of LLMs has surged, the rise of open source frameworks and tools has made it easier for developers and researchers to build, customise, and fine-tune their own LLMs. These frameworks provide accessible and flexible solutions for anyone looking to harness the power of LLMs without needing proprietary access to models like GPT-3.